NORMALIZED BASE CHANGE OVER A ONE-DIMENSIONAL REGULAR BASE IS SMOOTH AT GENERIC FIBRE POINTS

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Editorial note. Counted span is the author's own proof environment, source lines 19374-19398, immediately following the statement at 19357-19372.

Editorial note. The proof invokes the general theorem twice by \ref; those are navigation pointers into the same chapter, and the smoothness locus transported here is the one displayed in the span.

Editorial note. The author's phrase 'in the proof of Theorem ... we have seen that' refers to a computation inside the cited theorem; it is retained verbatim.

Allowed standard prerequisites (editorial).

- Normalized base change with reduced fibre, general form. chapter theorem theorem-normalized-base-change-with-reduced-fibre, cited at proof lines 19379 and 19394. Supplies, over each affine chart of S, the finite extension L_i/K and the smoothness statement whose locus is transported to the global situation.
- Normalization of a scheme in a finite field extension. standard Stacks construction used at proof lines 19369-19371 and 19381-19382. Defines T and the normalized base change Y over the integral closure B_i of A_i in L_i.
- Finite extensions of a field are directed and admit a common dominating extension. used at proof line 19383. Produces a single finite L/K dominating all the chartwise extensions L_i/K.

Source reference. Original theorem 09IL: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 19282–19282. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1 (Variant over curves)**.** *Let $f : X \to S$ be a flat, finite type morphism of schemes. Assume S is Nagata, integral with function field K, and regular of dimension* 1*. Then there exists a finite extension L/K such that in the diagram*

$$\begin{array}{ccccc} Y & \xrightarrow{\nu} & X \times_S T & \longrightarrow & X \\ & \searrow^{g} & \downarrow & & \downarrow f \\ & & T & \longrightarrow & S \end{array}$$

the morphism g is smooth at all generic points of fibres. Here T is the normalization of S in $\mathrm{Spec}(L)$ *and $\nu : Y \to X \times_S T$ is the normalization.*

Proof. Choose a finite affine open covering $S = \bigcup \mathrm{Spec}(A_i)$. Then K is equal to the fraction field of A_i for all i. Let $X_i = X \times_S \mathrm{Spec}(A_i)$. Choose L_i/K as in Theorem 09IL for the morphism $X_i \to \mathrm{Spec}(A_i)$. Let $B_i \subset L_i$ be the integral closure of A_i and let Y_i be the normalized base change of X to B_i. Let L/K be a finite extension

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dominating each L_i. Let $T_i \subset T$ be the inverse image of $\mathrm{Spec}(A_i)$. For each i we get a commutative diagram

$$\begin{array}{ccccc} g^{-1}(T_i) & \longrightarrow & Y_i & \longrightarrow & X \times_S \mathrm{Spec}(A_i) \\ \downarrow & & \downarrow & & \downarrow \\ T_i & \longrightarrow & \mathrm{Spec}(B_i) & \longrightarrow & \mathrm{Spec}(A_i) \end{array}$$

and in fact the left hand square is a normalized base change as discussed at the beginning of the section. In the proof of Theorem 09IL we have seen that the smooth locus of $Y \to T$ contains the inverse image in $g^{-1}(T_i)$ of the set of points where Y_i is smooth over B_i. This proves the lemma. $\square$